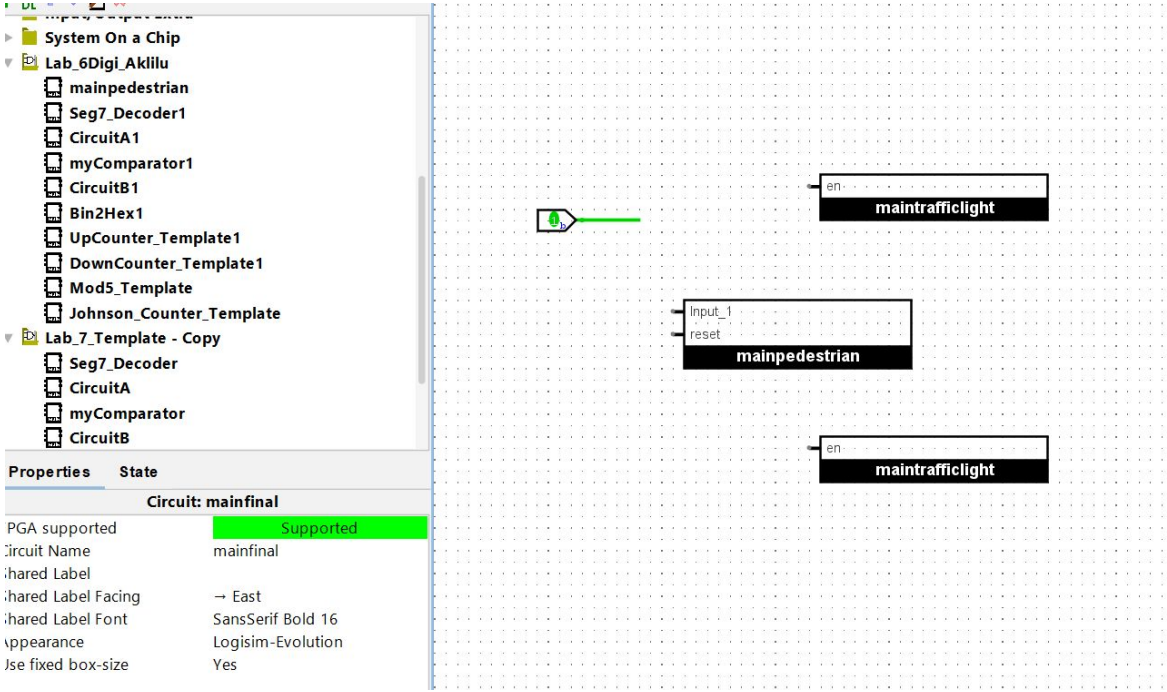
Final Project- Traffic light Intersection

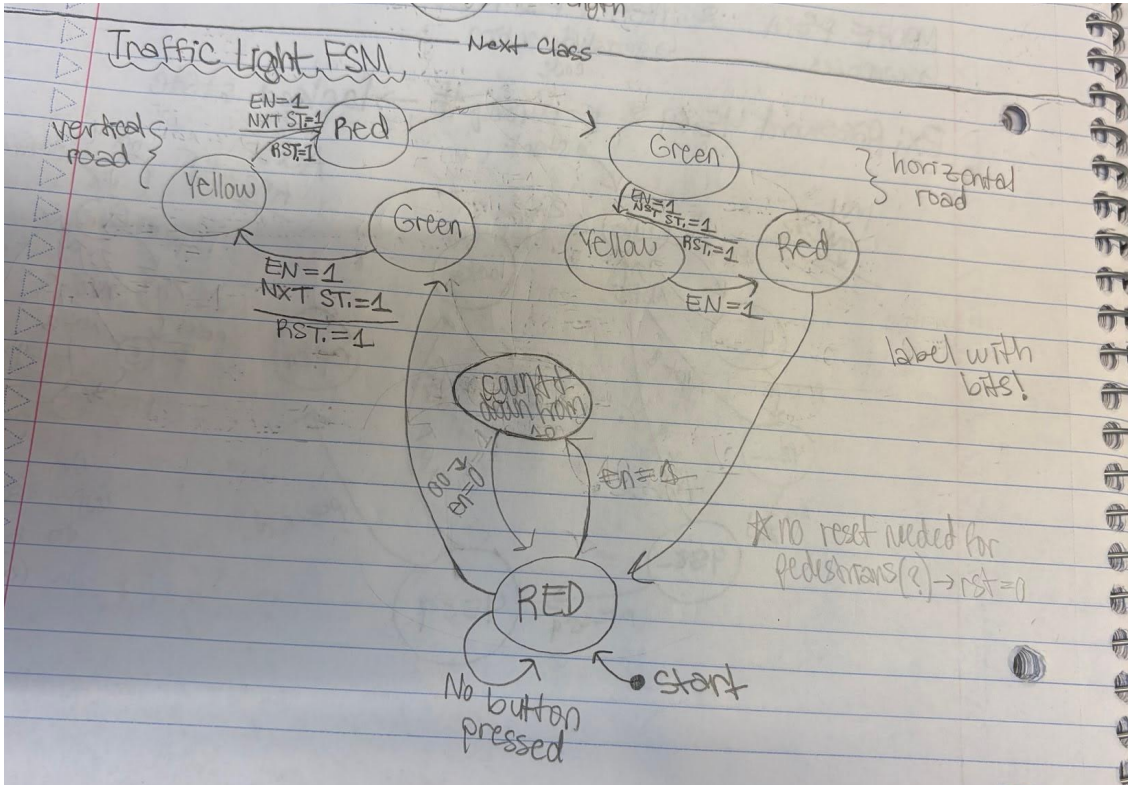
Eden Aklilu



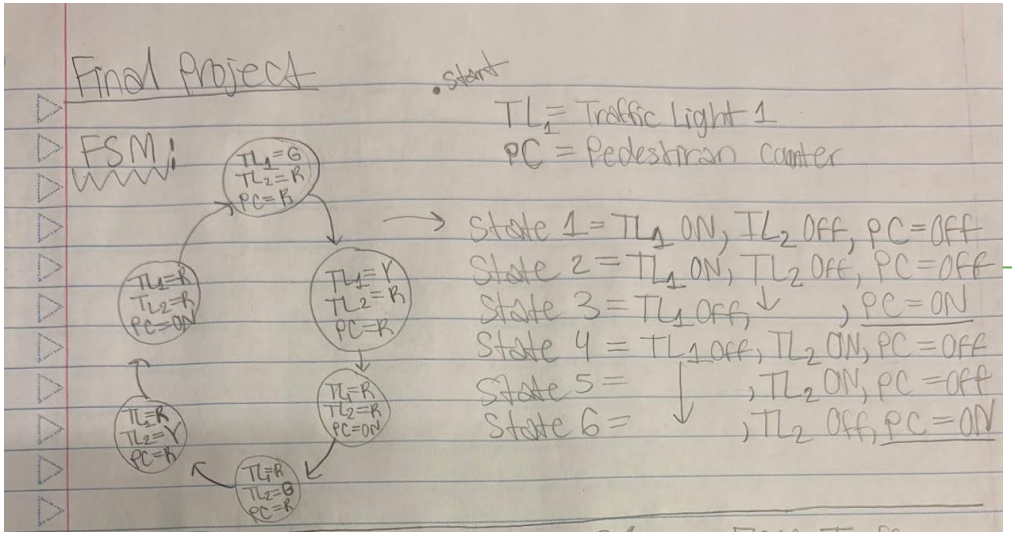
I began by loading the previous traffic light and pedestrian crossing counter projects into a new Logisim project to begin brainstorming an fsm

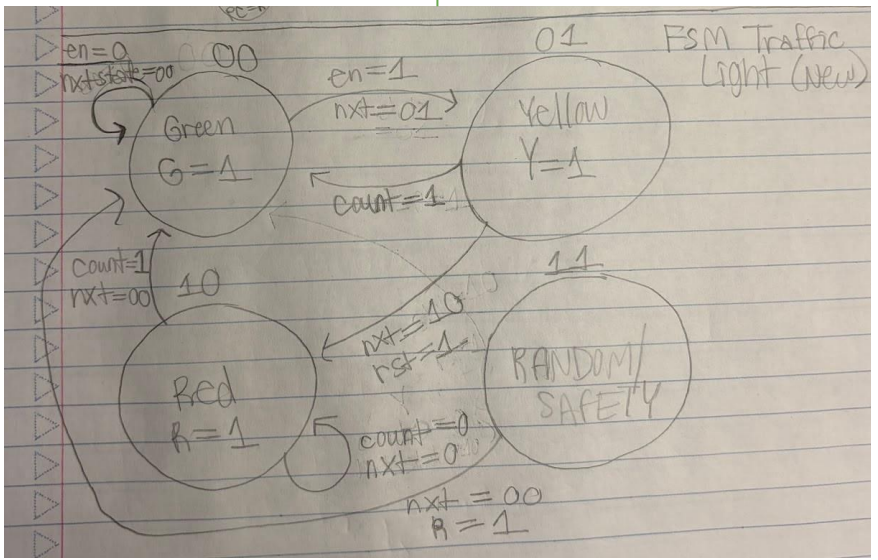


FSM chart draft 1 (unsure of how to connect everything together)



FSM draft 2 (with breakdown of states assisted by gemini)



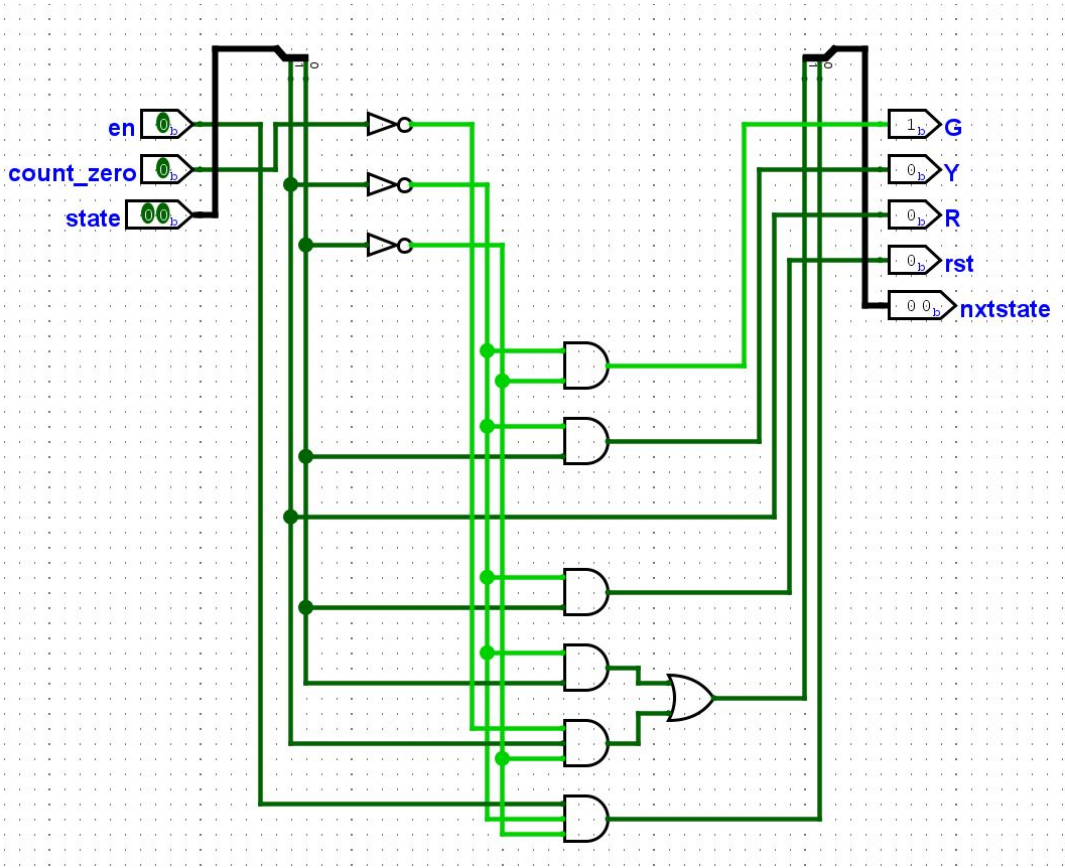


FSM Logic translated to traffic light fsm truth table

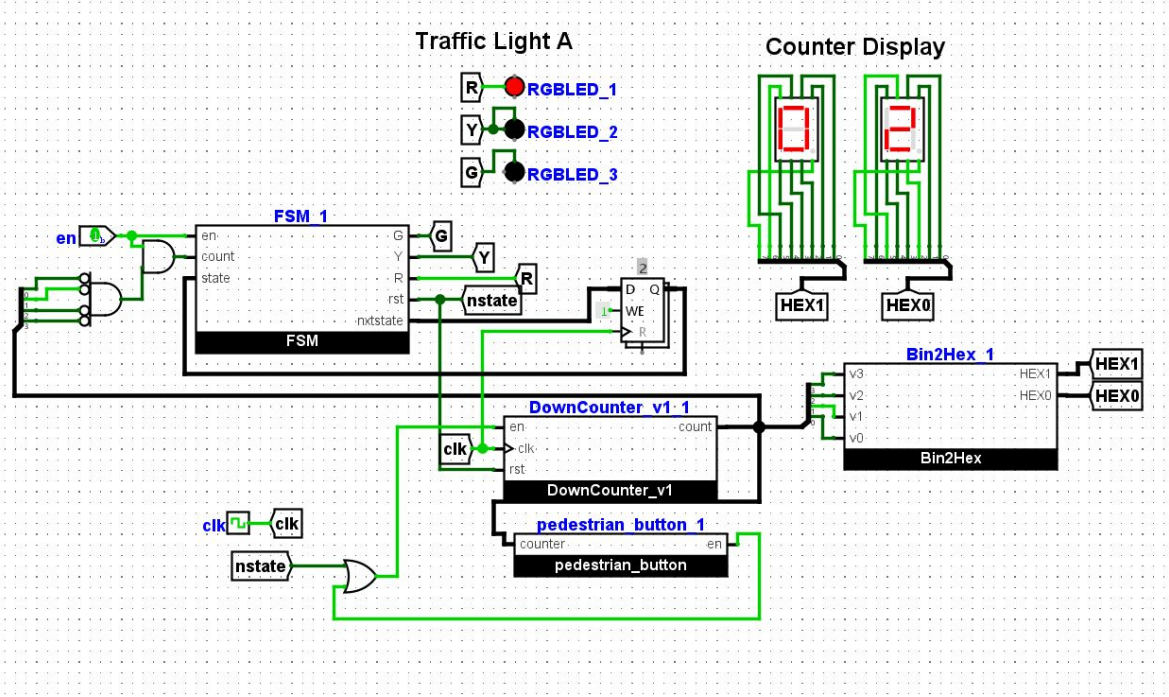
- 1 0 Collapse Duplicated Rows Show All Rows

16 of 16 rows shown

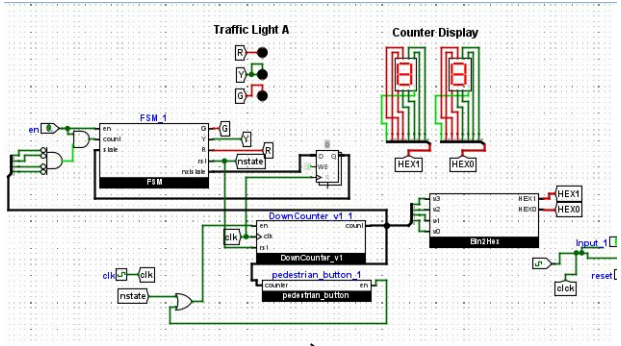
en	count_zero	state[1..0]	G	Y	R	rst	nxtstate[1..0]
0	0	0 0	1	0	0	0	0 0
0	0	0 1	0	1	0	1	1 0
0	0	1 0	0	0	1	0	1 0
0	0	1 1	0	0	1	0	0 0
0	1	0 0	1	0	0	0	0 0
0	1	0 1	0	1	0	1	1 0
0	1	1 0	0	0	1	0	0 0
0	1	1 1	0	0	1	0	0 0
1	0	0 0	1	0	0	0	0 1
1	0	0 1	0	1	0	1	1 0
1	0	1 0	0	0	1	0	1 0
1	0	1 1	0	0	1	0	0 0
1	1	0 0	1	0	0	0	0 1
1	1	0 1	0	1	0	1	1 0
1	1	1 0	0	0	1	0	0 0
1	1	1 1	0	0	1	0	0 0

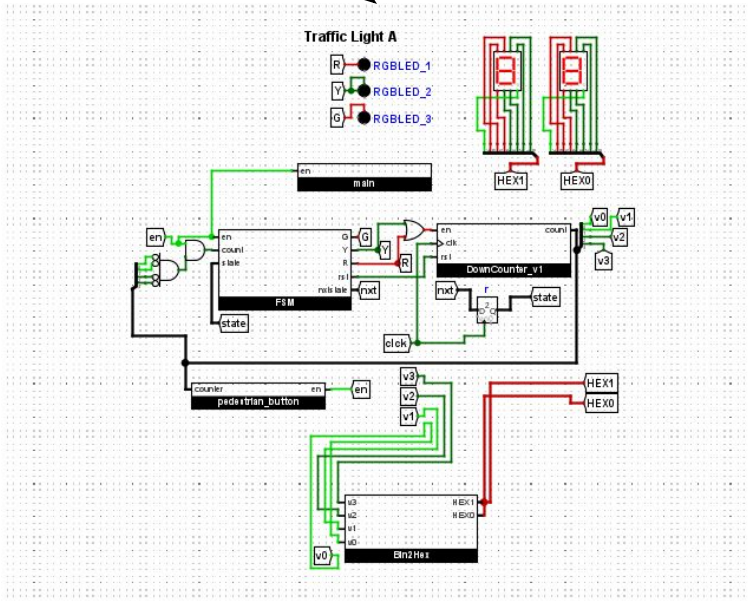


Added a register to the traffic light logic (ai helped me understand that the next state and state could be wired to the register as memory elements (as well as the clock for connection to the system)

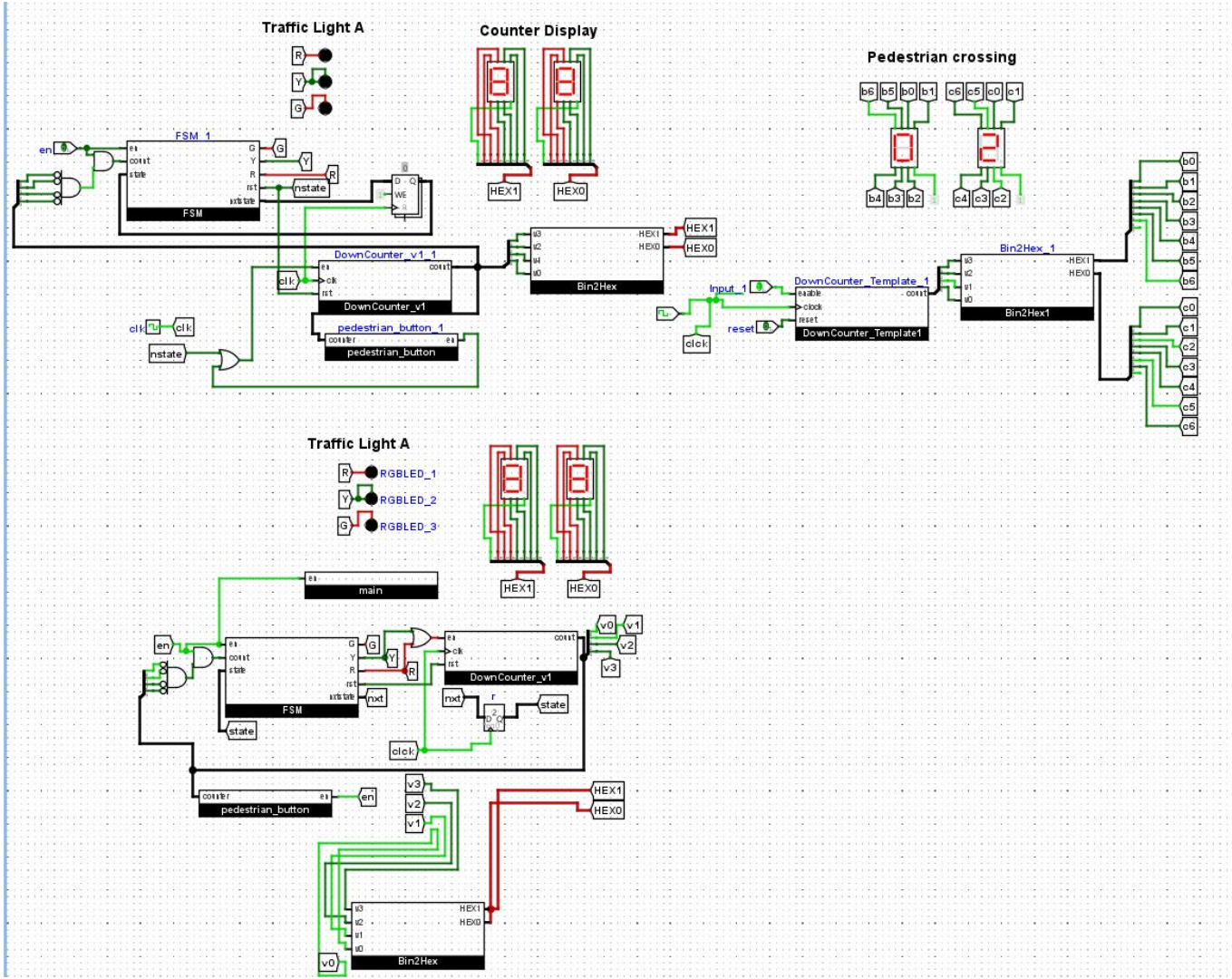


Putting everything together… (added register), attempt one and two at the traffic light- wasn't able to complete the project due to faulty wiring of some sort (even AI couldn't help with this one)





Final Schematic (unfinished product)



Reflection-
Project mostly done, I assume this would work on the FBGA had my wire connections worked out (specifically for next state and state inputs). I included all successful parts of this lab and am ready to take responsibility for tardiness and lack of ability to complete the project fully on time.